PS1 · Number system conversions (Ch 2) — Worked Solutions

Source: Digital Logic homework for Sep. 4 · study-slides (vetame21.github.io/study-slides)

> [ADDED] The original problem sets have no solutions. These worked solutions were written for this site; every conversion and every Boolean result was checked in code (truth tables and an exact K-map minimizer), and the VHDL was compiled. Compare them with your lecturer's model answers when those are released.

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1 Perform the following number system conversions.

(a) $1101011_2 = ?_{16}$

> **Answer:** $6\mathrm{B}_{16}$

Group 4 bits from the right: $110\ 1011 \rightarrow 0110\ 1011 = 6\ \mathrm{B}$.

Why this method: Binary ↔ hex: groups of 4 bits, because $16 = 2^4$.

(b) $174003_8 = ?_2$

> **Answer:** 1111100000000011_2

Replace each octal digit by 3 bits: $1 \rightarrow 001,\ 7 \rightarrow 111,\ 4 \rightarrow 100,\ 0 \rightarrow 000,\ 0 \rightarrow 000,\ 3 \rightarrow 011$, giving $001\ 111\ 100\ 000\ 000\ 011$. Drop the leading zeros.

Why this method: Octal → binary: one digit = 3 bits, because $8 = 2^3$.

(c) $10100.1101_2 = ?_{16}$

> **Answer:** $14.\mathrm{D}_{16}$

Group outward from the point. Whole part: $1\ 0100 \rightarrow 0001\ 0100 = 14$. Fraction: $1101 = \mathrm{D}$.

Why this method: Group from the radix point, padding with zeros at the outer ends.

(d) $\mathrm{F3A5}_{16} = ?_2$

> **Answer:** 1111001110100101_2

$\mathrm{F} \rightarrow 1111,\ 3 \rightarrow 0011,\ \mathrm{A} \rightarrow 1010,\ 5 \rightarrow 0101$.

Why this method: Hex → binary: one digit = 4 bits.

(e) $101111.0111_2 = ?_8$

> **Answer:** 57.34_8

Whole part: $101\ 111 = 5\ 7$. Fraction: $011\ \underline{100} = 3\ 4$ (pad with zeros on the RIGHT of a fraction). Check: $57.34_8 = 47 + \frac{3}{8} + \frac{4}{64} = 47.4375 = 101111.0111_2$.

Why this method: Binary → octal: groups of 3 bits from the point.

(f) $15\mathrm{C}.38_{16} = ?_2$

> **Answer:** 101011100.00111_2

$1 \rightarrow 0001,\ 5 \rightarrow 0101,\ \mathrm{C} \rightarrow 1100\ .\ 3 \rightarrow 0011,\ 8 \rightarrow 1000$, so $0001\ 0101\ 1100.0011\ 1000$. Drop the leading and trailing zeros.

Why this method: Hex → binary digit by digit.

2 Convert the following numbers into decimal.

(a) 1101011_2

> **Answer:** 107

$64 + 32 + 8 + 2 + 1 = 107.$

Why this method: Radix $r \to$ decimal: multiply each digit by its weight r^i and add.

(b) 174003_8

> **Answer:** 63491

$1 \cdot 8^5 + 7 \cdot 8^4 + 4 \cdot 8^3 + 0 + 0 + 3 = 32768 + 28672 + 2048 + 3 = 63491.$

Why this method: Weights are powers of 8.

(c) 10100.1101_2

> **Answer:** 20.8125

Whole: $16 + 4 = 20$. Fraction: $\frac{1}{2} + \frac{1}{4} + \frac{1}{16} = 0.8125$.

Why this method: Digits after the point have weights $2^{-1}, 2^{-2}, \ldots$

(d) $\mathrm{F3A5}_{16}$

> **Answer:** 62373

$15 \cdot 4096 + 3 \cdot 256 + 10 \cdot 16 + 5 = 61440 + 768 + 160 + 5 = 62373.$

Why this method: Weights are powers of 16; $\mathrm{A} = 10$, $\mathrm{F} = 15$.

(e) 12010_3

> **Answer:** 138

$1 \cdot 81 + 2 \cdot 27 + 0 \cdot 9 + 1 \cdot 3 + 0 = 81 + 54 + 3 = 138$. (The handout prints this as "120103": the subscript 3 was lost.)

Why this method: Same rule for any radix, here $r = 3$.

(f) $15\mathrm{C}.38_{16}$

> **Answer:** 348.21875

Whole: $1 \cdot 256 + 5 \cdot 16 + 12 = 348$. Fraction: $\frac{3}{16} + \frac{8}{256} = 0.1875 + 0.03125 = 0.21875$.

Why this method: Negative powers of 16 after the point.